

Jathurshan S.

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Summary

Electronics & Telecommunications Engineering undergraduate specializing in Machine Learning, with a focus on LLMs, Agentic AI, and ML for Systems (performance modeling). Proven ability to lead projects from conception to deployment, with hands-on experience in PyTorch, Transformers, and RAG

Education

University of Moratuwa

Moratuwa, Sri Lanka

2021 – 2025

- B.Sc. Eng. (Hons) in Electronics & Telecommunications Engineering; Minor in Mathematics
- **GPA: 3.61**; Dean's List Recipient (Semesters 1, 2, 6, 8)

Hartley College

Point Pedro, Sri Lanka

2011 – 2019

- G.C.E. Advanced Level — **Island Rank: 30** (3 A's), Z-Score: 2.8263

Technical Skills

- **Programming:** Python, C++, MySQL, JavaScript, SystemVerilog
- **ML/DL Frameworks:** PyTorch, TensorFlow, LibTorch, Transformers, scikit-learn, OpenCV, NumPy, Pandas
- **LLM / VLM / Agents:** LangChain, LangGraph, RAG, FAISS, Whisper, InternVL
- **Tools & Platforms:** Git, Github Actions, Docker, AWS (EC2, S3), Hugging Face, Anaconda, Streamlit, FastAPI, FFmpeg, MATLAB, Jupyter Notebook

Experience

Vega Innovations

Colombo, Sri Lanka

Nov 2023 – May 2024

- *Electronic Engineering Intern*
- Validated control pilot boards for Electric Vehicle (EV) charging stations, executing functional tests to ensure compliance with industry standards and interoperability.
- Designed and implemented a secure bootloader and memory access system, enabling robust over-the-air (OTA) firmware updates and signature verification.

Machine Learning/AI Projects

Story Generation Agent from Images (PyTorch, Transformers, LangChain)

[GitHub Repo](#)

- Integrated vision-language models (InternVL-1B) for event extraction and metadata creation from images.
- Utilized HuggingFace Transformers and LangChain to create an agent and generate engaging narratives based on extracted metadata.

Transformer-based Neural Machine Translation (PyTorch)

[GitHub Repo](#)

- Engineered a complete Transformer NMT pipeline with custom dataset preprocessing, causal masks, and positional encodings.
- Implemented encoder-decoder architecture with multi-head attention, feed-forward networks, and layer normalization following the original Vaswani et al. paper.
- Designed a modular training framework with PyTorch DataLoaders, optimized batching, and checkpointing.

Optimizing Tensor Program Execution in Domain Specific Languages (PyTorch, Adams2019)

[GitHub Repo](#)

- Generated the search space of possible schedules of programs from the intermediate representations.
- Developed a pre-trained model to predict the execution time of the program for the schedules.
- Integrated the cost model to guide the autoscheduling process.

Multimodal Document RAG Agent (PyTorch, Transformers, LangChain, FAISS) [GitHub Repo](#)

- Engineered a data pipeline to extract text and images from PDFs, generating unified vector embeddings for both using the CLIP model to create a persistent, searchable multimodal knowledge base with FAISS.
- Developed a Retrieval-Augmented Generation (RAG) agent using LangChain to dynamically retrieve relevant text and images, enabling a Vision-LLM to synthesize accurate, context-aware answers.

Video Subtitle Generator and Burner (OpenAI Whisper, moviepy, FFmpeg) [GitHub Repo](#)

- Implemented audio extraction from videos, AI-powered transcription to SRT format, and subtitle embedding using FFmpeg.
- Added efficiency features like skipping existing files to optimize batch processing.

Movie Recommendation System using LLaMA 3.1 (8B) (Python, LangChain, Streamlit)

- Built a movie recommendation chatbot powered by LLaMA 3 (8B), fine-tuned on movie metadata and user preferences.
- Integrated a retrieval-augmented generation pipeline using FAISS to provide context-aware and personalized recommendations.
- Developed a user interface for interaction using Streamlit on a local machine.

LSTM and GNN based Speedup Prediction (PyTorch, Hydra, NumPy) [GitHub Repo](#)

- Transformed high-level DSL code, including loops, computations, and memory access patterns, into a tree and graph representation.
- Trained a recursive LSTM architecture-based model and a GCN model to predict the speedup of the DSL programs by traversing through the program representation recursively.

RAG based Question Answering System (Python, LangChain)

- Built a Retrieval-Augmented Generation system using LangChain and PyTorch to answer questions from a document.
- Created a FAISS-based vector store for efficient retrieval of the documents.

Smartphone Image Super-Resolution (PyTorch) [GitHub Repo](#)

- Designed the generator and discriminator architectures of SRGAN.
- Created the loss functions for the generator and discriminator.
- Trained the model and obtained high resolution images.

Face Generation using VAE (PyTorch) [GitHub Repo](#)

- Implemented custom dataset from CelebA dataset for loading with data augmentation and preprocessing.
- Designed and trained a linear VAE model with encoder-decoder architecture for latent space representation.
- Defined custom loss function combining reconstruction and KL divergence terms.
- Created a complete pipeline from data loading to image generation, resulting in a reusable codebase.

CNN Image Classification (TensorFlow) [GitHub Repo](#)

- Implemented a Convolutional Neural Network (CNN) to classify images across multiple classes through supervised learning techniques.

Additional Projects

Processor Design

- Designed a non-pipelined RISC-V RV32I architecture processor using SystemVerilog. [GitHub Repo](#)
- Designed a pipelined RISC-V RV32I processor with instruction-level parallelism and hazard management. [GitHub Repo](#)
- Deployed both processor variants on the DE2-115 FPGA board for validation and performance testing.

Floating Point ALU Design

[GitHub Repo](#)

- Developed a 32-bit single-precision Floating Point ALU in SystemVerilog optimized for low resource utilization.
- Deployed, verified, and tested the ALU on the DE2-115 FPGA board.

Achievements

- **National Mathematics Olympiad:** Multiple-time medalist (Silver 2015–2016, Bronze 2014).
- **National Maths Quiz Competition:** Multiple-time medalist (Gold 2012, 2013; Silver 2014, 2015).
- **International Mathematics Competition:** Merit & Bronze (2014).

Reference

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